

Detailed Outline: Chapters 3–5

Measuring physics-driven and data-driven contributions
in hybrid process models

Chapter 3	Investigated Processes and Data Generation
Chapter 4	Results and Discussion (six sections)
Chapter 5	Conclusions and Outlook

Scope. This outline covers only what has actually been produced. Results are quoted where they exist so each section has something concrete to be built around. Items that are still missing are marked **GAP** and collected again at the end. Section numbering follows the six-section Chapter 4 agreed earlier; the overall discussion moves to Chapter 5.

Chapter 3. Investigated Processes and Data Generation

Two simulated processes. The first carries the main development; the second tests whether the methodology transfers. Chapter 2 stays general, so every process-specific choice belongs here.

3.1 Two-stage batch reactor

3.1.1 Process description

- Reaction network: stage 1 $A \rightarrow B \rightarrow C$; stage 2 $B + 2D \rightarrow E$ and $C + 2D \rightarrow F$.
- Competition for the shared reagent D — the mechanism that makes the misspecifications bite.
- Operating variables and their ranges; what is measured at each stage and when.

3.1.2 Correct and misspecified model variants

- Seven candidates M0–M6: correct; no side reaction; first order in D; wrong activation energy E_a ; lumped E/F; no D dependence; the source paper's own misspecification.
- For each: the modified equation, and what kind of error it represents (missing pathway, wrong order, wrong parameter, lumping).
 - State which are structural and which parametric — this distinction reappears in 4.4.

3.1.3 Kinetic parameter estimation

- **GAP in the current outline.** The mechanistic block is built from parameters *fitted to calibration data*, not literature values. This is load-bearing: the fitted-versus-nominal choice changed results materially.
- Objective, solver, bounds, starting values, convergence settings; fitted on calibration only.
- Report the fitted parameters per candidate and the residual cost as a mechanism-quality ranking.

3.1.4 Data generation

- SDE formulation, integrator, step size, diffusion coefficients.
- Discretisation and measurement noise; noise scale fixed from calibration batches only.
- **Dataset design** — currently missing from the outline and required by 4.5:
 - calibration, $T_2 \sim U(330, 370)$ K, $n = 100$
 - in-domain control, same distribution, independent draw, $n = 100$
 - extrapolation, $T_2 \sim U(370, 382)$ K, $n = 100$
- Why an in-domain control is necessary: without one, degradation cannot be separated from baseline difficulty. An earlier control drawn only from the top of the calibration range made extrapolation appear to *outperform* in-domain prediction.

3.1.5 Data blocks

- X_1 , M_2 , X_2 , Y : contents, dimensions, physical meaning.
- Stage-2 initial conditions for the mechanistic block come from upstream measured quantities, **not** from the first sample of the trajectory being predicted. The alternative makes the measured block a copy of M and collapses the iterative procedure of 4.5.
- Trajectory unfolding; truncation to a common length; the response taken at each batch's own endpoint.
- Note the structural ceiling this creates: the columns Y is an exact function of are not in X_2 .

3.2 Urethane manufacturing process

3.2.1 Process description

- Semi-batch reactor, two feed vessels; $A + B \rightarrow C$, $A + C \rightarrow D$, $3A \rightarrow E$.
- Competition for the shared reagent A — the analogue of D in 3.1.
- Manipulated inputs f_{v1} , f_{v2} , T ; measured species n_C , n_D , n_E .

3.2.2 Correct and misspecified model variants

- **GAP in the current outline** — 3.2 has no counterpart to 3.1.2, so the case study has no candidate mechanisms and cannot carry the methodology.
- Seven candidates U0–U6: correct reversible; irreversible second reaction (the source papers' own model); no isocyanurate reaction; first order in A; lumped rate constant; constant volume; temperature-independent kinetics.

3.2.3 Data generation and provenance

- Ground-truth and first-principles models; the mismatch is exactly the missing reverse step.
- **Provenance matters here** — the case study is assembled from three sources:
 - the seven initial molar charges are not tabulated anywhere; they are recovered by inverting the seven design variables of the original optimum-experimental-design paper
 - the control profiles are read from a figure, with a stated reading uncertainty
 - the two source papers conflict on whether the stated noise is a variance or a standard deviation; the conflict is resolved by a signal-to-noise argument
- Design: 50 calibration + 10 validation batches, 90 h, sampling every 30 min.
- **GAP:** there is no extrapolation design for this process — the published validation batches are drawn from the same input ranges as calibration. Either design one, or state explicitly that case study 2 tests the calibration-side diagnostic only.

3.2.4 Data blocks

- M, X, Y; the KD inputs are moved into M by the causal-ordering rule.
- The mechanistic block spans the full batch while the measured block stops at a decision point — the single-decision-point form of the procedure in 2.6.
- **Response selection.** Final n_C is unusable: its measurement noise exceeds its entire batch-to-batch spread (SNR 0.37–0.99), and every model returns negative Q^2 . n_D has SNR ≈ 143 and is the species the structural mismatch acts on.

3.3 Verification of the generated data

- **GAP in the current outline.** This is what makes both case studies credible and currently appears nowhere.
- Case study 1: reproduction of the reference latent-variable allocations {4, 6, 2} and {3, 3, 6}.
- Case study 2, three independent checks:
 - atom balances on A and B close to machine precision
 - the reverse rate constant implied by the equilibrium relation agrees with the value another study discovered from data alone, to 0.03–0.07 standard deviations
 - the mechanistic mean absolute error for species C reproduces the published value to within 1.5 % in calibration and 5 % in validation
- Where replication is not exact, say why — the two unstated choices are the residual weighting in parameter estimation and whether time enters the data-driven model.

source: regen_corrected_data.py, urethane_case_study.py, urethane_replicate.py

Chapter 4. Results and Discussion

Six sections. The arc: do the methods work → how is information split → is M2M stable → does the modification help → does it predict extrapolation → does it transfer.

4.1 Baseline multiblock model comparison

Purpose: establish that all three methods are adequate here, and justify SO-PLS as the vehicle.

- MB-PLS, SO-PLS, SMB-PLS; latent-variable selection by RMSECV; reproduction of the reference allocations.
- Comparable predictive accuracy across methods; SO-PLS adopted for its variance-decomposition property, which the other two do not admit.
- Remark: SMB-PLS reduces to SO-PLS on this data — the block score is exactly parallel to the super-score, so the three-way shared/unique split is degenerate. Derivation to an appendix.

source: `sopls_two_orderings.py`, `sopls_two_block.py`

4.2 Decomposition of mechanistic and measured contributions

Purpose: this is the motivation for the modified ratio in 2.4, and should say so explicitly.

- Sequential sums of squares under both block orderings and the resulting asymmetry.
 - for the correct model, SS_M falls from 71.72 to 16.43 when M is placed last, while SS_X rises from 5.71 to 61.58
 - roughly 56–67 % of SS_Y is jointly explainable and unattributable by any single ordering
- Unique versus shared contributions via block removal; the commonality identity $\text{Joint} = \text{Unique}_M + \text{Unique}_X + \text{Shared}$.
 - the two unique parts capture only 17–37 % of the joint effect; Shared is 0.47–0.68 out of a joint of about 0.84
 - the data-driven baseline is identical across candidates because it contains no mechanistic block — note what this implies for cross-model comparison
- Species-level removal: which species carry the contribution.
- Protocol sensitivity: re-optimising the sub-models versus holding them at the full model's allocation roughly doubles the unique mechanistic contribution. State which is used.

source: `sopls_commonality.py`, `X2_removed_analysis.py`, `shared_variance_forward_reverse.py`

4.3 Sensitivity of M2M to latent-variable allocation

Purpose: show the point estimate is not trustworthy, and define what replaces it.

- Måge plot and the full allocation grid; M2M varies by orders of magnitude across allocations of near-identical RMSECV.
- Definition and size of the 1-SE flat region; region sizes and the standard error per model.
- Distribution of M2M within the region; regional median versus point estimate; the optimum is not a representative allocation.
- Why the rule stops at one standard error:
 - the first standard error buys simplification at about 0.42 percentage points of Q^2 per latent variable removed; the second costs about 1.33 — three times the price for less than half the benefit
 - at two standard errors the internal Q^2 spread of the region roughly doubles, so the premise that members are equivalent no longer holds
 - state the cost honestly: the 1-SE rule is not free, it costs 1.8–3.2 standard deviations of partition noise
- Stability across cross-validation partitions.

source: `lv_allocation_analysis.py`, `metric_ratio_lv_distribution.py`, `metric_ratio_flat_regions.py`, `q2_flat_region_comparison.py`, `sopls_two_block_ceiling.py`

4.4 Original versus modified M2M ratio

Purpose: the contribution. Report what it does and what it does not do.

- Both ratios across the seven candidates; SS- and Q²-based estimators compared.
- The reference value 1: which mechanisms fall either side and how decisively.
 - mechanisms omitting a whole reaction pathway fall below 1; the structurally sound ones rise above it, unanimously across cross-validation partitions
- Where the modification does not help:
 - it does not separate the correct model from the wrong-activation-energy variant
 - it does not rank mechanisms by correctness — what it measures is how heavily the hybrid leans on the mechanistic block
- Numerical behaviour: unbounded above, undefined when the denominator reaches zero. Report the fraction of allocations where this occurs and how the flat region suppresses it.

source: `metric_unique_m2m.py`, `M2M_vs_AMR_shared_comparison.py`

4.5 Extrapolation validation

Purpose: does a calibration-only diagnostic predict out-of-domain behaviour?

- Datasets and the in-domain control; what “extrapolation” means here.
- **4.5.1 Single-decision-point test** — full measured trajectory available.
- **4.5.2 Iterative (rolling) prediction** — at each decision point the unobserved part of the trajectory is replaced by mechanistic predictions.
 - report performance against decision point for both regimes
 - misspecified mechanisms recover as observations accumulate; sound ones are stable
- **4.5.3 Diagnostics versus extrapolation performance** — rank correlation across the candidates, for the point estimate and the regional median.
 - the regional median outperforms the point estimate
 - calibration fit alone predicts nothing, so the diagnostic adds real information
- **The two evaluation modes disagree**, and the reason is identifiable: at the fully observed endpoint the mechanistic block is redundant, so a model leaning on it looks marginally worse. Only the rolling regime tests what the metric is about. Report the single-decision-point result as a negative control that motivates the rolling design.

source: `extrapolation_test.py`, `extrapolation_1se_distribution.py`, `regen_fit_and_test.py`,
`regen_fit_and_test_paperIC.py`, `iterative_prediction.py`

4.6 Application to the urethane manufacturing process

Purpose: does the methodology transfer? Partly — and the failure is informative.

- **4.6.1 Replication of published results** — parameter estimates and error metrics against the source study; where agreement is exact and where it is not.
- **4.6.2 M2M analysis** across the seven candidate mechanisms, at several decision points.
- **4.6.3 Degeneracy of the modified ratio.** The measured block adds almost nothing beyond the mechanistic block, so the denominator collapses and the ratio becomes undefined — including for the correct mechanism, the worst place to lose it.
 - the original SS-based ratio remains defined and still ranks mechanism quality
- **Scope condition.** The determining factor is whether the mechanistic block can forecast beyond the measurement window. In case study 1 it cannot, and the unique contribution of the measured block is healthy; here it can, and that contribution vanishes. State this as a condition of applicability rather than as a limitation.

source: `urethane_case_study.py`, `urethane_replicate.py`, `urethane_m2m.py`

Chapter 5. Conclusions and Outlook

5.1 Summary of findings

- What the original M2M ratio does and does not measure.
- What the modified ratio adds: an intrinsic reference point, and no credit for shared variance.
- What it costs: a denominator that can vanish.

5.2 Methodological recommendations

- Report the ratio as a regional median with an interval, not as a point estimate.
- Report the shared component alongside it, so the reader sees what the ratio does not account for.
- Pin down the sub-model allocation protocol in the definition.
- Evaluate extrapolation with the rolling procedure; the fully observed endpoint is not a test of mechanistic contribution.

5.3 Scope conditions and limitations

- The metric applies where the measured block retains a non-trivial unique contribution.
- Rank correlations over a handful of candidate models have very low power; say so.
- Results rest on single realisations unless repeated across generation seeds.
- Absolute values are not comparable across data regimes; only within-regime comparisons hold.

5.4 Outlook

- Repeat across generation seeds to confirm the extrapolation result.
- Re-estimate mechanistic parameters on data spanning the extrapolation range, to separate “wrong mechanism” from “parameters estimated in the wrong domain”.
- Investigate the fraction of allocations at which the measured block adds nothing out-of-sample, which separated sound from broken mechanisms more cleanly than the ratio itself and was not pursued.
- Bounded reformulations of the ratio that avoid the vanishing denominator.

Open decisions and gaps

Collected from above. Each needs either work or an explicit statement in the text.

Section	Gap	Action
3.1.3 / 3.2.3	Parameter estimation has no subsection	Write it; state the objective, solver and that fitting uses calibration only
3.1.4	Dataset split not described	Add calibration / in-domain control / extrapolation
3.2.2	No candidate mechanisms for urethane	Add U0-U6, mirroring 3.1.2
3.2.3	No extrapolation design for urethane	Design one, or state that case study 2 is calibration-only
3.3	No verification section	Add; the replication checks already exist
4.4	Protocol for sub-model allocation	Fix one convention and state it
4.5	Single seed only	Repeat across generation seeds, or state as a limitation
Ch. 2	LV selection rule stated inconsistently	Argmin or parsimony within 1-SE - decide once

One caution on reuse. Several numbers quoted in this outline come from specific data regimes that are not interchangeable — real versus regenerated data, fitted versus nominal kinetic parameters. Before a number goes into the thesis, check which regime produced it against the source file named under each section.